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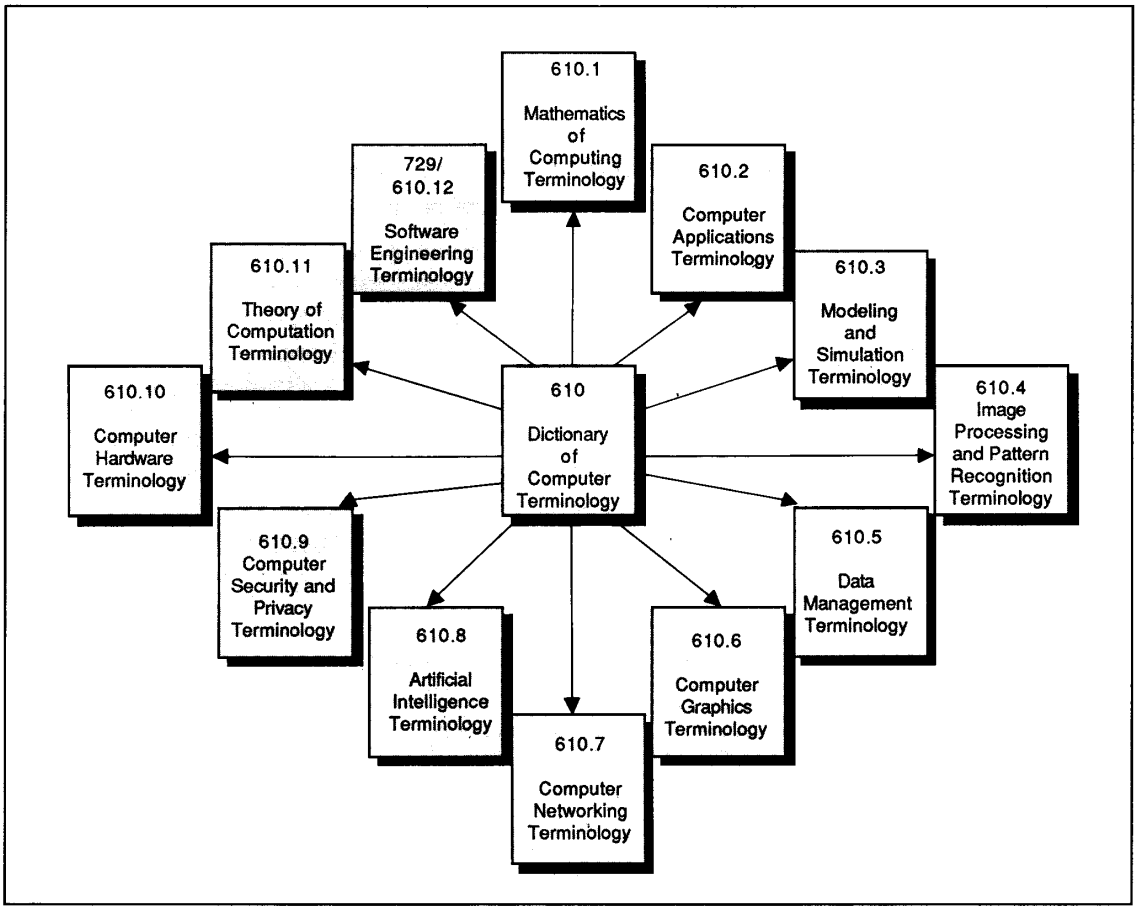

Figure 1. Planned products of Project 610.

books, journal articles, dictionaries, and whatever other sources we can find. To date, the working group has gathered a total of 185 computer-related glossaries and dictionaries, plus hundreds of nondictionary sources. We make it a point to find each term in multiple sources to achieve a balance of ideas. On the average, we look up each term in five sources. We then select the best definition or, more often, write a composite definition reflecting the best elements from several definitions or descriptions. In those cases where we quote a definition verbatim, we make note of the source(s) quoted.

Refining definitions for consistency. It came as a big surprise to us that drafting the definitions is only about 20 percent of the job. The bigger job comes after all of the terms have draft definitions. It consists of making the definitions work together. Consider the following definitions for "constant" and "variable":

- constant: a data item whose value does not change
- variable: a quantity that can assume any of a given set of values

Taken individually, each of these definitions isn't bad. Taken together, they don't work very well. Why is a constant a "data item," when a variable is a "quantity"? Why are the last parts of the two definitions not more parallel to emphasize the contrast? The definitions should start out using identical words and remain as parallel as possible so that the contrast comes across clearly. For example,

- constant: a data item whose value cannot change
- variable: a data item whose value can change

The English language is full of synonyms and near-synonyms that make it possible to say the same thing in many different ways. A major challenge that the working group faces is to use consistent language throughout the dictionary so that the reader never falsely assumes that we chose two different words to mean two different things when in fact the terms we chose are simply synonyms.

The key mechanism for achieving this consistency is our "divide and conquer" approach, described above. By grouping each term with logically related terms, both the definer and the reviewer can ensure that related terms are defined consistently. Our coordination and consistency subgroup, described below, also addresses this issue.

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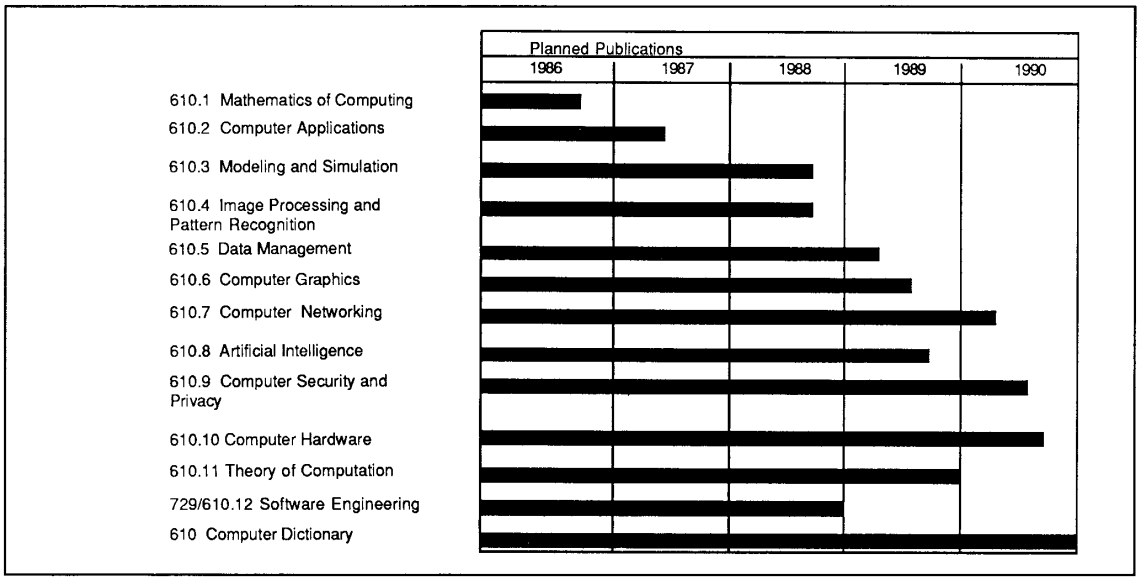

Figure 2. Planned schedule for Project 610 glossaries.

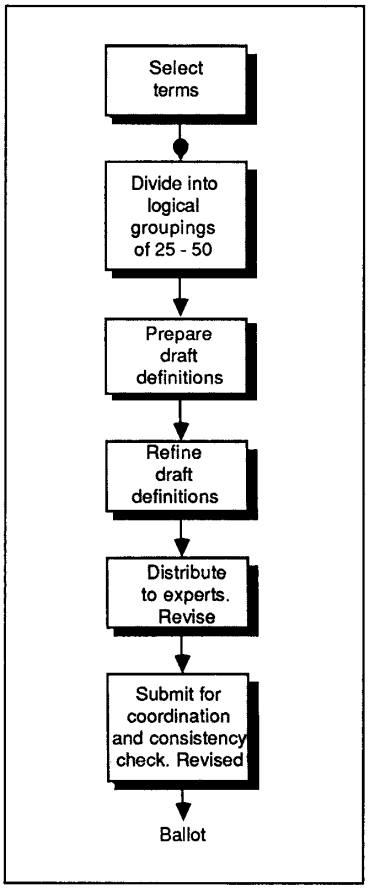

Figure 3. The Project 610 process.

Expert review. The working group has taken great pains to recruit subject-matter experts to review the definitions and tell us whether they are complete and correct. The members who do the defining may not be experts in a particular field and may have to rely on reading textbooks and articles to prepare the definitions. We want to be sure that experts in each field agree with our work. Each set of definitions is sent out at least once, often two or more times, to experts for review. The definitions are then revised to incorporate any comments received.

Coordination and consistency review. The final step for each set of definitions is a review by our coordination and consistency subgroup. This group is responsible for ensuring that all definitions follow our style guide and are consistent with one another. Any problems found are corrected before IEEE balloting.

Special challenges. The dictionary effort faces some special challenges that make it interesting and unique.

Breadth of scope. Most IEEE working groups focus on one particular topic—how to write a software quality assurance plan, or what the pin assignments should be on a particular bus. Such working groups are composed of experts and interested parties in that particular field. Our scope includes the entire spectrum of computer-related topics.

One of our special challenges has been to identify and recruit definers and reviewers who together offer expertise in all of these fields. To meet this challenge, our membership chairman, Mary Yee, has made hundreds of telephone calls to universities and companies, written articles for computer-related publications, and prepared flyers for distribution at conferences. The success of our effort is largely due to her recruiting efforts.

Size of the dictionary. Most IEEE standards are no more than 35 pages long. These standards typically take three years to develop. The computer dictionary will be approximately 400 pages long. It is, in essence, twelve simultaneous efforts rolled into one. The logistics of such an effort are considerable. We try to keep all twelve areas moving forward at a steady pace, but some move quickly and others languish for distressingly long periods. As on all projects, volunteers come and go, and have varying degrees of time, energy, and interest to contribute.

The original plan called for the dictionary to be developed, balloted, and published as one entity. It soon became clear, however, that the rate of completion of the different subject areas was going to differ dramatically—by years in some cases—and it was decided to ballot and publish the completed subject areas as separate glossaries as we went along. This approach has two benefits. It is a great morale booster for the working group to see a glossary

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published, and it provides the fruits of our work to interested parties.

Issues in dictionary development. We have been continually surprised by the number and type of issues that arise in what would appear to be a straightforward process. The paragraphs below provide a sample of the questions we have addressed.

Who is our audience? In preparing the dictionary, should we assume the reader has detailed knowledge of computers, no knowledge of computers, some technical knowledge, no technical knowledge, or something in between? The answer is critical because it influences our selection of terms and the wording of our definitions.

After much discussion, we decided that we are writing the dictionary for any person who will come across computer-related words in his or her reading, either on or off the job. That mental picture of our reader has led us to maintain a balance between the technical precision needed by experts and the clarity needed by laymen.

How should terms be organized? The answer to this question may seem obvious—the terms should of course be organized from A to Z. One of the basic issues that we faced at the beginning of our project, however, was a strong minority position that the dictionary should be organized by topic, with terms organized from A to Z under each topic.

There is precedent for such an organization. The *Data Processing Vocabulary of the International Standards Organization (ISO)* is organized in this manner. The argument for this arrangement is that it places terms in context and is more informative than an overall list. The disadvantage is that when you pick up the dictionary, you can't just go to the term based on its spelling. You have to know what section the term is in, or look it up in an index in the back of the dictionary.

After much debate, we decided that the glossaries and the overall dictionary would be more usable with an overall A to Z organization in each. We try to ensure that each definition depicts the context in which the term is used.

Which alphabetization scheme should we use? Even the decision to order the terms from A to Z was not the end of the story. There are two different ways to alphabetize when a dictionary includes both words and phrases. Consider the terms "adding" and "add transaction." Which would you put first? The two schools of thought here are whether or not to ignore the blank. If you ignore the blank, then "add transaction" is alphabetized as though it were "addtransaction" and it goes after "adding." If you take the blank into account, then "add transaction" goes before "adding."

Corresponding questions arise on how to alphabetize hyphens and slashes. Does "line-end zone" go before or after "line filling"? Does "input/output" go before or after "inputs"? Various dictionaries take different stands on these questions.

We decided to take blanks into account and to regard hyphens and slashes right after blanks in alphabetical order. This rule seemed to us to put the terms in the order that most readers would look for them.

Should the dictionary be descriptive or prescriptive? Dictionary writers face the question "Should we tell it like it is or tell it like it should be?" The official terms for these choices are *descriptive*, meaning describing current usage, versus *prescriptive*, meaning prescribing how terms should be used.

There are advantages and disadvantages to both approaches. To be completely descriptive, we would need to seek out and reflect every way in which each term is used. This approach is informative, but seems to contradict the notion of developing a standard dictionary. To be completely prescriptive establishes a standard, but often ignores reality.

An example of an area in which this issue arises is the terminology related to fault tolerance. Experts in fault tolerance have carefully thought through the difference between a human mistake, the resulting fault in a computer program, and resulting failure when the program is run, and the amount by which the output is in error. Most people call each of these an error—"I made an error; my program contains an error; my program generated an error; the result is in error by five miles." The fault tolerance folks, however, carefully distinguish between mistakes, faults, failures, and errors.

The question for the dictionary project is whether to reflect the way most people talk about these concepts or to reflect the careful, but not widely used, terminology of the fault tolerance field. After much debate on this issue, we chose a middle ground. We are including in the dictionary the most common current usage(s) of each term and also, with explanations, prescribed but uncommon usages.

Should abbreviations be included? We computer types use a lot of abbreviations. The working group decided early in the project to include abbreviations. The harder question was whether to intermix the abbreviations with the other terms or to present them in a separate listing at the back of each glossary.

The deciding factor was again usability. Several members pointed out that terms like Basic and Cobol are in fact abbreviations, but the reader might not know that and would look in the main dictionary, not in the abbreviation list. Furthermore, these terms have in some sense gone beyond being abbreviations and are terms in their own right. Once we saw that we could not clearly distinguish between terms and abbreviations, we decided to make one overall list containing both types of entries.

Should we include proprietary terms? A major source of debate has been

Current leaders in IEEE Project 610

The following list shows Project 610 leaders' names in bold, their affiliations in parentheses, and their specializations within the 610 group:

Jane Radatz (Logicon, Inc.)—overall 610 project, software engineering

Anne Geraci (Burroughs Wellcome)—computer applications, modeling and simulation, image processing and pattern recognition, data management

John Lane (Edinboro University of Pennsylvania)—coordination and consistency

Louise McMonegal (US Naval Facilities)—computer graphics

Russell Mikel (IIT Research Institute)—artificial intelligence

Larry Reeker (BDM Corp.)—theory of computation

Charles Russell (Logicon, Inc.)—computer security and privacy

Ernest Stalder (US Dept. of the Interior)—computer networking

Michael Strait (University of Maryland)—computer hardware

Mary Yee (Logicon, Inc.)—membership and publicity

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Areas addressed by 610 subgroups

Computer hardware
- networking hardware
- IC technology
- computer architecture
- analog computer hardware
- mainframe hardware
- digital logic components
- microprocessors
- microprogramming

Software engineering
- addressing
- assembling, compiling, linking, loading
- cohesion and coupling
- computer performance evaluation
- configuration management
- data types
- debugging
- documentation techniques
- documentation types
- errors, faults, and failures
- evaluation activities
- formal reviews and audits
- instruction formats
- language types
- libraries
- microprogramming
- module interaction
- operating system concepts
  - exception/interrupt handling
  - job/supervisor concepts
  - job scheduling methods
  - resource management
- path analysis and proof of correctness
- program statements/instructions/operations
- programming languages
- quality attributes
- requirement types
- software and system testing
- software architecture
- software development process
- software development techniques
- software miscellaneous
- software tools, utilities

Mathematics of computing
- arithmetic errors
- basic math
- Boolean algebra
- complementation
- computer arithmetic
- error detection and correction
- mathematical notation
- number conversions
- number systems
- numeric codes
- numerical analysis
- punched card formats
- shifts
- statistics and probability

Computer graphics
- general
- graphics hardware concepts
- graphics equipment
- graphics techniques
- graphics attributes
- graphics primitives
- graphics inputs
- 3D graphics

Theory of computation
- automata
- Petri nets
- computational complexity
- Chomsky hierarchy
- phrase structure grammars
- formal languages
- parsing
- strings
- models of computation
- Turing machines
- program logics
- program semantics
- computability
- recursion

Artificial intelligence
- subareas of AI
- graphs
- rule-based programming
- AI theory
- logic
- knowledge representation
- expert systems
- computer learning
- natural language processing
- AI programming languages
- inference and control of reasoning
- searching
- robotics
- computer vision

Computer networking
- networks
- data communications
- general communications
- error conditions
- protocols
- signal processing
- open systems architecture
- local area networks
- networking hardware
- telephone networks

Computer security and privacy
- administrative issues
- communications security
- emanations
- encryption
- formal verification methods
- general
- threats
- physical security
- piracy
- privacy
- passwords
- security assessment
- security-related hardware
- software measures
- access concepts
- types of security

Data management
- characters
- data
  - attributes
  - data elements
  - fields
  - files
  - general
  - records
- data dictionary
- data structures
  - linear
  - nonlinear
  - trees
- data types
- database—general
- keys
- searching and searching algorithms
- sorting
  - general
  - algorithms
- data access methods

Image processing and pattern recognition
- pattern recognition
- images
- digital images
- image processing techniques

Modeling and simulation
- types of models
- types of simulations
- simulation—general
- game theory
- queueing theory
- variables

Computer applications
- automated language processing
- automatic indexing
- business data processing
- CAD/CAM
- character recognition
- computer-assisted instruction
- control systems
- critical path method
- library automation
- medical applications
- micrographics
- miscellaneous
- office automation
- operations research
- personal computing
- robotics
- scientific and engineering applications
- telecommunication applications
- types of processing
- user concepts
- word processing
  - general
  - text editing
  - text formatting

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whether to include terms like AT&T's Unix and IBM's SNA, which are widely used but are, in fact, product names. The argument in favor of including them is that people will run across them and look them up. The argument against them is that they are commercial products and, since we cannot possibly include all competing products, we should not include any. In this case, we decided in favor of caution over helpfulness. Proprietary terms will not be included. We still have mixed feelings about some of our favorite terms, but decided on the safer course.

How much tutorial information should definitions include? Some of our members are uncomfortable limiting the entry for each term to a simple definition. They advocate a more encyclopedia-like approach, with background and tutorial information provided for each term. We decided not to take this approach. Our selected format includes a definition, cross references to related terms, and, in some cases, examples and brief notes. We feel that we serve our readers best with this format.

How to handle definitions from other sources? One of our key operating procedures is to keep careful track of sources and to give credit when we quote a definition verbatim. Questions began to arise right away, however, over this seemingly straightforward policy. The thing about definitions is that every word and every comma count. When we find a definition that we like, but want to delete one comma, are we still quoting the definition? How about when we change "the" to "a" or "which" to "that"? How about when we change "see ..." to "see also ..." On one hand, we want to be fair and give credit where credit is due. On the other hand, we aren't sure how a source might feel if one of its definitions is changed, but we still credit the source as though it were quoted exactly.

Our final decision was to keep track internally of whether a definition is quoted exactly, slightly adapted, or changed so much that it could no longer be considered quoted. In the final product, we cite the source only for definitions that we quote exactly. We believe that this is the fairest approach.

Dare we disagree with other standards? The other side of avoiding plagiarism is deciding whether to cite, without question, definitions that appear in IEEE and other standards. After all, the argument goes, if the definitions have already been standardized, who are we to alter the wording?

This approach seems good in theory, but we soon found that definitions drawn from different sources have inconsistent styles, use words and concepts differently, disagree among themselves, and become confusing and inappropriate when placed in a new context. To deal with this problem, we have adopted an approach of quoting standard definitions whenever possible, but maintaining the freedom to change the style and wording of these definitions if they are incompatible with others in the dictionary.

Should we avoid quoting? The working group has been advised not to quote any definition except those in existing IEEE standards, to avoid copyright problems. We balked at this advice. It seems to us that if our overall goal is to promote standardization, then to change a perfectly good definition is counterproductive. So, while fewer than 10 percent of our definitions have ended up being direct quotes, we have reserved the right to quote other sources, cite these sources, and obtain copyright permission when we wished to use a definition verbatim.

Should we write in American or British English? The IEEE is an international organization. While we are not equipped to define the terms in the languages of all countries having IEEE membership, we wanted at the least not to exhibit American parochialism. To this end, we took pains to recruit reviewers from Canada, Great Britain, and other English-speaking countries and to reflect alternate spellings (such as center and centre) and usage in the dictionary.

How does our effort relate to a similar effort by X3K5? Accredited Standards Committee X3K5 is charged with developing and updating a dictionary of information processing terminology that somewhat overlaps the scope of Project 610. X3K5 published the *American National Dictionary of Information Processing* (ANDIP) in 1977 and an updated version called the *American National Dictionary of Information Processing Systems* (ANDIPS) in 1982. It is currently working on an update to ANDIPS.

An ongoing issue is how our two groups can best cooperate. To promote this cooperation while striving to serve our somewhat diverse constituencies, project 610 uses ANDIPS as one of its sources, encourages X3K5 to use 610 as one of its sources, and welcomes the continued participation of a liaison representative from X3K5.

Looking ahead. 610 is an interesting and challenging project. Those of us involved have become close friends with large vocabularies. The end is still a long way off, but our progress is steady and we are determined to complete the job.

Our work will never be completely finished, however. By the time we publish the entire dictionary, it will be time to start revising. We try not to think about that too much.

Volunteers are always welcome. We need definers and reviewers, both for the subject matter areas and for the coordination and consistency subgroup. If you would like to contribute in any of these ways, please contact our membership chair, Mary Yee, at (202) 646-2222.

Finally, I would like to thank our former and current sponsors in the Computer Society, Herb Hecht, Fletcher Buckley, Laurel Kaleda, and Helen Wood; our advisor from the IEEE Standards Office, Frank Jay; our IEEE editors, Judy Gorman and Sandra Valentin; those who have sent us dictionaries and good wishes; and all members of the 610 working group. These individuals have offered support, encouragement, and help to our project. We are grateful to each of them.

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Chairperson

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